Supplement: Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials

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Contents

This document is a companion to the manuscript "Latent-class and mixture-model formula-

tions for biomarker-treatment interaction in N-of-1 trials" (pmsimstats team). The main manuscript was compressed to a twelve-page working length by moving the extended literature review, the full finite-mixture taxonomy, the full biomarker-moderation spectrum, and two secondary theoretical asides here. Section numbering, notation, and citation keys match the main manuscript; this supplement does not stand on its own and should be read alongside it.

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1 Extended literature review

The main manuscript's introduction summarizes, in a single paragraph, the two methodological traditions from which the present paper draws: the psychometric tradition, which developed the continuous-versus- discrete question in its own right beginning with Spearman and Lazarsfeld, and the econometric and marketing-science tradition, which developed the covariate-dependent gating function that the biomarker- moderation problem requires. We set out that history in full here.

1.1 Psychometric origins

The continuous-trait view in formal statistical work originates with Spearman's two-factor theory of intelligence[1], in which performance on cognitive tests was modeled as the sum of a single general factor and test-specific factors. Spearman's machinery was elaborated through the 1930s and 1940s by Thurstone, Hotelling, Lawley, and others into the modern factor-analytic tradition, in which observed indicator variables are written as linear combinations of unobserved continuous factors plus residual noise[2,3]. Within this tradition, between-person heterogeneity in any latent construct is by assumption continuous, and individual differences are summarized by factor scores along one or more dimensions.

The discrete-class view emerged from a parallel but methodologically distinct literature beginning with Lazarsfeld's work on latent structure analysis in the 1950s[4,5]. Lazarsfeld observed that in attitudinal and survey data, the joint distribution of categorical responses to multiple items frequently displayed structure inconsistent with a single continuous latent trait, and instead suggested that respondents belonged to one of several qualitatively distinct latent classes within which item responses were locally independent. This 'local independence' formulation is the founding assumption of latent class analysis. Goodman[6] placed the LCA model on rigorous statistical footing, deriving identifiability conditions and maximum-likelihood estimation procedures for the binary-indicator case in a Biometrika paper that remains a standard reference. Through the 1970s and 1980s, the LCA framework expanded to polytomous and continuous indicators, the latter widely known as latent profile analysis[7], to multilevel and longitudinal extensions[8], and to mixtures with covariate-dependent class probabilities[9,10]. McLachlan and Peel[11] unified much of this material into a comprehensive monograph on finite mixture models that is the standard entry point for quantitative researchers from outside the discipline.

The defining tension in psychometric methodology has been whether a given pattern of observed heterogeneity is best understood as continuous or as discrete. Bartholomew[12] framed this as the choice between 'graded' and 'unfolding' models of individual differences and argued that the data themselves rarely settle the question without strong prior assumptions. The empirical literature has accumulated examples in both directions: cognitive ability is generally well-described by continuous factor models, while certain forms of psychopathology display latent-class structure consistent with qualitatively distinct subtypes. Subsequent methodological work, beginning with[13] and[14], developed hybrid frameworks (factor mixture models) in which discrete classes coexist with within-class continuous factors, taken up in more detail below.

1.2 Econometric and marketing-science development

A largely independent strand of work on finite mixtures developed in econometrics and quantitative marketing through the 1980s and 1990s. The motivating problems differed from psychometrics: rather than measurement of cognitive or psychological constructs, the econometric concern was unobserved heterogeneity in micro-econometric models (consumer choice, labor-market participation, firm productivity) where standard regression assumptions of identical agents were known to fail. Heckman and Singer[15], in an Econometrica paper on duration models, demonstrated that estimates of structural parameters in survival-style models could be highly sensitive to the assumed functional form of the unobserved heterogeneity distribution, and recommended a non-parametric maximum-likelihood approach in which the heterogeneity distribution itself was estimated as a discrete mixture with unknown support points and weights. Their estimator anticipated much of what would later become standard finite-mixture methodology in biostatistics, and Lindsay[16] placed the broader theory of mixture likelihoods on a unified mathematical footing.

In quantitative marketing, the parallel concern was customer segmentation. A population of consumers might respond to a marketing intervention differently depending on which unobserved segment they belonged to, and the firm needed both to estimate segment-specific response functions and to predict segment membership for new consumers. DeSarbo and Cron[17] formalized this as 'clusterwise linear regression', in which observations were partitioned into clusters with cluster-specific regression coefficients. Wedel and DeSarbo[18] generalized the approach to generalized linear models, and Jedidi and colleagues[19] extended it to structural equation models with unobserved segments. Within this literature, the gating function (the model for the probability of segment membership as a function of covariates) became a first-class object, and methodological work focused on whether the gating model could share covariates with the within-segment regression model. Jacobs and colleagues[20], working contemporaneously in machine learning, formalized an essentially identical structure under the name 'mixtures of experts', which provided much of the now-standard vocabulary.

Two themes from the econometric strand are particularly relevant to the N-of-1 trial setting. First, the econometric tradition treated unobserved heterogeneity as a nuisance in many applications, and was correspondingly attentive to whether the heterogeneity distribution was identified, weakly identified, or over-fit. The identifiability concerns raised by[15] and elaborated by[16] translate directly into N-of-1 settings where sample sizes are small and the heterogeneity is the principal object of inference rather than a nuisance. Second, the marketing-science focus on covariate-dependent gating functions corresponds exactly to the biomarker-as-class-membership-predictor structure that motivates the present paper. The biomarker-treatment moderation question, translated into econometric vocabulary, is whether the gating function depends on a single observed covariate (the biomarker) and whether the within- class regression coefficient on treatment differs across classes.

1.3 Convergence on hybrid frameworks

By the late 1990s, both the psychometric and the econometric strands had converged on a hybrid formulation: combine continuous within-class factor structure with discrete between-class structure. Yung[13], in Psychometrika, developed the factor mixture model as a finite mixture of confirmatory factor models, where each class had its own factor-loading and residual covariance structure. Arminger and colleagues[14] placed factor mixture models in the broader Mplus-style structural equation framework. Lubke and Muthen[21] conducted the first systematic Monte Carlo evaluation of factor mixture models and clarified the conditions under which the discrete-class component was empirically distinguishable from continuous heterogeneity. Clark and colleagues[22] surveyed the modeling strategies and presented an applied analysis of psychiatric comorbidity.

In parallel, the longitudinal extension of these ideas yielded the growth mixture model[23–25] and the closely related group-based trajectory model[26,27]. Both posit that a longitudinal outcome trajectory is generated by a finite mixture of class-specific trajectories. Growth mixture models permit within-class random-effects variation around the class-specific mean trajectory; group-based trajectory models constrain within-class variance to zero, so each class has a single deterministic trajectory. Nagin's group-based approach was developed primarily in criminology to characterize developmental trajectories of antisocial behavior, and has been widely applied in epidemiology, public health, and clinical research where trajectory heterogeneity is the principal scientific question.

The methodological optimism around growth and factor mixture models during the early 2000s was tempered by a series of papers documenting overextraction hazards. Bauer and Curran[28], in Psychological Methods, demonstrated through simulation that growth mixture models fitted to data generated from a single non-gaussian distribution can spuriously identify multiple latent classes; the apparent classes are artifacts of distributional non-normality rather than evidence of substantive subpopulation structure. Bauer[29] elaborated on the implications for substantive psychological research and recommended caution in interpreting class structure without independent validation. Lubke and Muthen[21] documented the analogous concerns for factor mixture models, identifying the joint conditions on sample size, class separation, and class-specific parameter differences under which class structure is recoverable. Nylund and colleagues[30] systematised the class-enumeration question through Monte Carlo evaluation of information criteria and likelihood-ratio bootstrap procedures. The cumulative message of this literature, referenced repeatedly in the main manuscript, is that mixture models can extract apparent class structure where none exists, and that distinguishing genuine from spurious class structure requires either substantial samples (typically several hundred to several thousand participants) or strong prior information.

1.4 Translation into longitudinal biostatistics

Mixture and latent-class methods entered biostatistical practice through several complementary channels. Verbeke and Lesaffre[31], in the Journal of the American Statistical Association, proposed a linear mixed-effects model in which the random-effects population was itself a finite mixture of multivariate normals. Their motivating application was longitudinal HIV-infection data in which latent classes of patients displayed qualitatively different CD4 trajectories. The model is essentially a growth mixture model in biostatistical clothing, with the standard random-intercept-and-slope linear mixed model augmented by a discrete latent class indicator. Zhang and Davidian[32] extended the framework to allow more flexible random-effects distributions via semi-parametric mixture specifications.

Proust-Lima and colleagues developed a more comprehensive software framework with the `lcmm` R package[33], which implements joint latent-class mixed models for longitudinal continuous and ordinal outcomes, with optional joint modeling of survival outcomes via shared latent class structure. The `lcmm` framework has been deployed in cognitive aging research, Alzheimer's disease progression modeling, and clinical-trial latent-class discovery.

The `flexmix` package in R[34,35] provides a more general infrastructure for fitting mixtures of regression-style models with user-supplied component specifications, and has been used in pharmacokinetic and pharmacodynamic modeling where individual-level variation in dose-response curves motivates a mixture structure.

Despite this translation activity, mixture and latent-class methods have not, to the best of our knowledge, been systematically applied to aggregated N-of-1 trial designs. The published N-of-1 methods literature is dominated by within-participant effect estimation[36,37], Bayesian hierarchical models for participant-specific effects[38], and aggregated-evidence meta-analytic frameworks[39,40]. The biomarker-moderation question has received occasional treatment under the heading of precision-medicine power calculation[41,42], but in each case the moderation has been specified as a continuous interaction term in a fixed-effects analysis, with no parallel treatment of the latent-class formulation. The mixture-formulation question appears to be unaddressed in the existing N-of-1 methodology literature, which is the gap the main manuscript takes up.

2 Finite-mixture taxonomy in full

The main manuscript's overview table locates four families of finite mixture model from the psychometric and econometric literatures that bear directly on the biomarker-treatment interaction problem. We develop each family, its origin, and its specific relevance to the N-of-1 trial setting here.

2.1 Latent class and latent profile analysis

Latent class analysis and latent profile analysis are mixture models for cross-sectional data. Each participant belongs to one of a small number of unobserved classes; the observed variables have class-specific marginal distributions. LCA uses categorical indicators; LPA uses continuous indicators. A mixture fitted to baseline biomarkers alone returns posterior class-membership probabilities that can be used as covariates in subsequent outcome analysis, the two-stage analytic strategy standard in psychometric applications. In the biomarker-trial context, an LPA-style model posits two or more responder classes with class-specific marginal distributions of on-drug and off-drug response, and estimates class-membership probabilities from the observed response trajectories rather than from baseline covariates alone. Classical references include[5],[6],[43], and[11]; the Vermunt and Magidson Latent GOLD software and the `poLCA` R package implement the standard EM estimation routines.

2.2 Growth mixture models and group-based trajectory models

Growth mixture models[23–25] extend LPA to longitudinal data by specifying class-specific growth curves (or, in the trial setting, drug-response curves) with within-class random-effects variation. A GMM is a longitudinal linear mixed model in which the population distribution of random effects is a mixture of multivariate normals, each corresponding to a distinct trajectory class. Group-based trajectory models[26,27] are the degenerate case in which within-class random-effect variances are set to zero, so that participants within a class share an identical trajectory up to observation-level residual noise.

GMM is the closest psychometric analogue of the faithful latent-class generative form developed in the main manuscript: each participant belongs to a latent class with its own drug-response trajectory, and the class label is drug-state-invariant. A GMM that uses the biomarker as a class-membership predictor is essentially that generative model. Bauer and Curran[28] and Bauer[29] document the identifiability and overextraction hazards that accompany GMM estimation at realistic sample sizes, showing in particular that GMM can extract apparent class structure from data generated by single-class non-gaussian processes. These cautions apply with full force to aggregated N-of-1 trials, where participant counts are typically well below the sample sizes at which GMM class-count recovery has been validated.

2.3 Factor mixture models

Factor mixture models[13,14,21] combine continuous latent factors with discrete latent classes, yielding a generative structure in which within-class variation is governed by a factor model and between-class variation is governed by class-specific means and, optionally, class-specific factor loadings or residual covariances. The FMM framework subsumes both the pure continuous case (single-class factor model) and the pure discrete case (LCA with no within-class structure) as limits.

FMMs are two-level hierarchies. At the upper level, participants are assigned to one of a small number of classes, as in LCA. At the lower level, each class has its own within-class random-effects structure. The biological hypothesis the FMM expresses is that there are genuinely distinct responder classes and that, within class, additional continuous heterogeneity is driven by covariates or unmeasured individual differences. For the biomarker-moderation problem, FMM provides a principled way to represent a biomarker as both a noisy class indicator (via the gating function) and a continuous modulator of within-class response (via factor loadings on the response factor). The hybrid representation matches more biological scenarios than either pure LCA or pure factor analysis, at the cost of correspondingly higher estimation requirements.

2.4 Regression mixtures and mixtures of experts

Regression mixtures[17–19] and the closely related mixture-of-experts architecture[20] allow class-specific regression coefficients with class-membership probabilities (the gating function) that depend on covariates. An ordinary linear model with a biomarker-treatment interaction posits a single regression surface in which the treatment slope varies smoothly with the biomarker. A regression mixture instead posits two or more distinct regression surfaces, one per class, with a separate covariate-dependent model governing which surface applies to which participant. The two-stage structure (classification by the gating model, then prediction by the within-class regression) parallels the two-stage structure of causal-inference methods that use propensity scores to stratify before estimating within-stratum treatment effects.

In the trial-simulation idiom, regression mixtures correspond to a DGP in which the biomarker governs both the probability of being a responder and the magnitude of the within-responder drug effect. The `flexmix` package in R[34,35] implements this family with user-definable component models, including mixed-effects components suitable for repeated-measures data.

3 Biomarker-moderation spectrum in full

The main manuscript locates the covariance-only formulation of[42] within a broader spectrum of biomarker-moderated DGPs, organized moment by moment (biomarker moderating the mean, the covariance, both, or higher moments). The full development of each point on the spectrum is given here.

3.1 Covariance-only moderation

The covariance-only specification holds $E[BR_{it}]$ fixed across biomarker strata and places the interaction entirely in $\mathrm{Cov}(B_i, BR_{it} \mid D_{bc,it})$. High- and low-biomarker participants have the same expected drug response at any given time but differ in the covariance between biomarker and response, specifically in how strongly the biomarker co-varies with response during on-drug versus off-drug periods.

This is an unusual restriction within the psychometric mixture tradition. Most standard FMM and GMM parameterizations permit class-specific means at a minimum, and the means-constant-covariance- varies case is typically encountered only as a constrained submodel used for testing measurement-invariance hypotheses[44,45]. The corresponding biostatistical model has two latent classes with identical expected outcomes but differing residual variance or within-subject autocorrelation; such models are occasionally fitted as sensitivity analyses but rarely assumed as the primary DGP. The covariance-only encoding's appeal lies in its tractability for power simulation under multivariate normality: the entire DGP reduces to drawing from a single MVN with a constructed covariance. Its cost is that it does not correspond cleanly to any natural generative mechanism.

3.2 Combined mean and covariance moderation

Arminger and colleagues[14] develop finite mixtures of conditional mean- and covariance-structure models in which both the expected-response vector and the residual covariance matrix differ between classes and may depend on observed covariates within class. This is the general psychometric form: it reduces to a mean-moderation model in the limit of a single class with covariate-dependent mean only, and to the covariance-only encoding in the limit of class-invariant means with covariate-dependent covariance only.

In the combined form, responders and non-responders differ both in typical drug response (mean moderation) and in how noisily that response is expressed within class (covariance moderation). Both channels carry information about class membership: the mean channel via $E[BR \mid B]$, the covariance channel via $\mathrm{Var}[BR \mid B]$. Under carryover, the mean channel is comparatively robust because class-conditional means need not converge off-drug; the covariance channel is eroded by construction because off-drug covariance is attenuated. A combined-specification simulation should therefore exhibit power loss intermediate between the pure-mean and pure-covariance extremes; this is plausibly the most biologically faithful regime for biomarker-guided psychiatric trials. Verbeke and Lesaffre[31] provide the analogous treatment for linear mixed-effects models with heterogeneity in the random-effects population, which is the more direct generalization of the repeated-measures structure used in N-of-1 trial simulation.

3.3 Location-scale moderation

Generalised additive models for location, scale, and shape[46] and the related distributional regression tradition[47] allow the biomarker to enter multiple parameters of the outcome distribution simultaneously through separate link functions. In plain terms, a standard regression models the conditional mean of an outcome as a function of covariates; a GAMLSS-style distributional regression models the conditional mean, the conditional variance, and if appropriate the conditional skewness and kurtosis, each as a separate function of covariates. The familiar special case is heteroscedastic linear regression with covariate-dependent residual variance; GAMLSS generalizes this to arbitrary distributional shape.

In the biomarker-trial context, a GAMLSS-based DGP would allow high-biomarker participants to have both a shifted mean drug response (a location effect, which is the mean-moderation architecture) and altered response variability (a scale effect, analogous to the covariance-moderation architecture's covariance moderation but operating on marginal rather than joint moments), without requiring discrete class structure. The GAMLSS framework is thus the continuous-heterogeneity analogue of FMM and occupies an intermediate position between the pure mean-moderation architecture (mean only) and the pure covariance-moderation architecture (covariance only). This intermediate position is attractive for sensitivity analysis because it does not require committing to a specific number of latent classes.

3.4 Heterogeneous random-effects models

Verbeke and Lesaffre[31], Zhang and Davidian[32], and Proust-Lima and colleagues[33] develop linear mixed-effects models in which the random-effects distribution is itself a finite mixture, allowing class-specific random-intercept and random-slope distributions. The biostatistical setting this extends is the standard linear mixed model with random intercept and random treatment slope, familiar from any longitudinal trial analysis using `nlme::lme` or `lme4::lmer`. In the standard model, the random treatment slope is drawn from a single normal distribution with mean equal to the average treatment effect and variance capturing individual heterogeneity. In the heterogeneous random-effects extension, that single distribution is replaced by a mixture: a fraction of participants are drawn from a 'responder' distribution with a large mean treatment slope and another fraction from a 'non-responder' distribution with a near-zero mean treatment slope. The biomarker predicts which mixture component a participant comes from.

Applied to aggregated N-of-1 trial data, this framework represents responder heterogeneity directly at the participant-specific random treatment effect, which is arguably the most clinically transparent representation available: responders and non-responders have distinct random-slope distributions on the treatment indicator, and the biomarker predicts class membership rather than entering the analysis-model mean structure at the fixed-effect level. The `lcmm` package[33] implements this family and is the closest available tooling for the longitudinal aggregated N-of-1 use case: it accepts standard longitudinal mixed-model syntax, fits a mixture on the random-effects distribution, and returns posterior class probabilities for each participant. This is the DGP family used for the Study 5 pilot reported in the main manuscript, and is the lowest-friction entry point for an extension of the existing aggregated N-of-1 simulation framework.

4 Software for mixture analyses, discussion

The main manuscript's software table lists the R packages relevant to the models surveyed above, in approximate order of fit to the longitudinal aggregated N-of-1 use case. We elaborate briefly on two entries here. The `lcmm` package[33] is used for the Study 5 pilot in the main manuscript because its syntax is closely related to `nlme::lme` with a latent-class extension, which keeps the class-aware analysis wrapper close to the existing `lme_analysis()` contract. The `flexmix` package[34,35] is more general, supporting arbitrary user-defined component models including mixed-effects components, but requires substantially more configuration and is not used in the pilot reported here.

The reference implementation outside R is Mplus[48], which remains the standard environment for GMM, FMM, and related mixture-SEM specifications. Much of the published psychometric simulation literature on mixture-model identifiability and overextraction hazards[21,28,30] uses Mplus, and direct comparison with that literature occasionally requires access to it.

5 Secondary theoretical notes

Two points relevant to the paper's argument are developed here rather than in the main text, because neither is required to follow the paper's central claim.

5.1 Why the analysis model is simpler than the data-generating process

A natural reaction to the simulation program described in the main manuscript is that the DGP is structured (a three-component decomposition into biological response, placebo-belief response, and time-variant natural history under the standard pmsimstats parameterization; or a latent-class mixture under the alternatives in this paper) while the analysis model is a four-term linear mixed model `Sx ~ bm + t + Dbc + bm:Dbc` with random intercept and AR(1) residual correlation. A reader newly encountering this asymmetry is reasonable to ask why the analysis does not mirror the DGP component-by-component, and whether doing so would improve inference about the moderation estimand.

A component-matched analysis (a non-linear mixed-effects model fitting Gompertz BR, PB, and TV components with participant-specific random effects) is impractical for trial-relevant N for four reasons: the trial design must supply the contrasts that identify each component (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV), and many designs supply only some of them; each Gompertz component costs roughly three to four parameters at the population level and additional random-effect parameters per participant, which at $N \in [30, 150]$ inflates the variance of the moderation estimand more than it reduces residual confounding; the non-linear fitting machinery has higher convergence-failure rates and is more sensitive to starting values than the linear-mixed analogue; and the moderation estimand $\beta_{bm:D}$ is robust to the BR-versus-PB attribution for the validation question, even though it is not robust to the larger questions of whether the moderation is class-structured (the main manuscript) or covariance-versus-mean-encoded (the `dgp-mean-moderation-vs-mvn` companion).

The smallest extension that retains tractability while improving the BR-versus-PB sensitivity of the inference is a phase indicator plus its interaction with the drug indicator, schematically `Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase + (1 | ptID)`. This adds three to four parameters and tests two things directly: whether the apparent drug effect shrinks under blinding (a signature that the open-label drug effect was partly PB-mediated) and whether the moderation itself shrinks under blinding (a signature that the biomarker-treatment interaction was partly biomarker-by-PB rather than biomarker-by-BR). For papers where pharmacological-versus-placebo disentanglement is a secondary scientific question, this extension is the right step; for papers where the primary estimand is the marginal moderation, the simple `Sx ~ bm + t + Dbc + bm:Dbc` analysis is adequate. The simulation program fixed in the companion plan uses the simple form throughout; the phase-indicator extension is a candidate sensitivity analysis to add to Study 1 (the mixture-vs-MVN factorial) once the primary cells are run. The document at `docs/24-component-decomposition-pedagogy.md` develops this point in more detail with worked numerical examples.

5.2 The latent-class DGP as a generator of apparent subadditivity

A connection worth flagging explicitly: the latent-class DGP specified in Study 1 is a natural mechanism for generating apparent subadditivity in a downstream one-component analysis, even when no direct $BR \times PB$ interaction is specified in the class-conditional generative model. The companion methods paper at `analysis/report/06-component-decomposition/` defines subadditivity as a non-zero coefficient γ on a $BR \times PB$ interaction term in the conditional-mean specification of the additive decomposition, and notes that $\gamma > 0$ in a fitted additive-extension analysis is ambiguous between two distinct underlying mechanisms.

The first mechanism is direct mechanistic interaction: shared neural substrate, ceiling effects on the symptom scale, or belief-dynamics adaptation each produce a true non-additive coupling between drug and placebo response. The second mechanism is emergent subadditivity from latent-class structure: if the trial population is a mixture of responder and non-responder classes (the formulation in the main manuscript) and class membership correlates with both

BR and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the additive-extension analysis this looks exactly like a γ-weighted interaction.

The Study 1 pre-registration accordingly includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels: a PB-class-independent cell (matching the original specification) and a PB-class-correlated cell (generating apparent subadditivity). The Study 1 simulation output therefore lets the analyst distinguish the two mechanisms empirically: if the additive-extension $\hat{\gamma}$ is large in the PB-class-correlated cell but small in the PB-class-independent cell at matched marginal effect size, the mechanism is emergent rather than mechanistic. The cross-paper synthesis with the BR-PB-TV decomposition paper is therefore not merely methodological convenience but a direct route to falsifying or confirming a class-aware versus mechanistic-interaction interpretation of any apparent subadditivity finding.

6 Simulation infrastructure and pre-flight pilot detail

The main manuscript reports the primary Study 5 result at 240 replicates per cell. This section records the infrastructure build-out and the earlier 100-replicate pre-flight pilot that motivated the choice of primary class-aware moderation test, for the reader who wants the full audit trail.

The five-study production program is pre-registered under the ADEMP framework of Morris, White, and Crowther[49] at `analysis/scripts/latent-class-mixture-application/00-ademp-pre-registration.md`. Phase 1 of the simulation program has reached the infrastructure-and-pilot milestone. ADEMP tables for all five studies are committed, with cell counts (approximately 2,300 cells across the program), seeded random-number control, an effect-size tuning sub-study, and a deviation-log specification. Subsequent production runs are diagnosed against these pre-registered estimands rather than against any post-hoc wrapper choice.

Class-aware analysis wrapper. An `lcmm_analysis()` wrapper around `lcmm::hlme` is committed at `01-lcmm-wrapper.R` with the same input-output contract as the existing `lme_analysis()`. The wrapper uses a two-stage fit (a single-class `hlme` is fitted first to anchor starting values for a `gridsearch`-based multi-start `hlme(ng=2)` fit) and extracts two moderation signals: the gating-coefficient z-test on the biomarker (the primary class-aware moderation test, identifying whether the biomarker predicts class membership) and the within-class $\beta_{bm:D}$ t-test (the secondary signal, retained for inferential symmetry with the linear-mixed baseline).

Heterogeneous-random-slopes DGP. A wrapper around `generateData()` that adds a Bernoulli class-membership channel with logistic gating in the standardized biomarker and class-specific multipliers on the BR factor at on-drug timepoints is committed at `02-heterogeneous-slopes-dgp.R`. On a single-replicate smoke test ($N = 70$, gating slope 1.5, responder multiplier 1.0, non-responder multiplier 0.0) the DGP produces 46% responders and a marginal $\mathrm{Cor}(B, BR) = 0.63$ that the linear-mixed `bm:Dbc` test detects ($p = 0.002$) without class awareness; the `lcmm` fit returns a posterior class-assignment entropy of 0.86 and a BIC 170 lower than the matched single-component MVN-DGP fit, indicating that the mixture model discriminates the two DGPs at this parameter cell.

Study 5 pre-flight pilot. A 100-replicate pilot under the single-component null DGP (the covariance-moderation architecture with $c_{bm} = 0$, $N = 70$, hybrid open-label (OL) design) is reported at `output/03-study5-pilot.rds`. Per-replicate timing is 0.05 s for the linear-mixed baseline and 1.57 s for the class-aware analysis (32:1 cost ratio, consistent with the smoke-test estimate). `lcmm` convergence rate is 96%; the four non-converging replicates encountered numerical issues during the gridsearch initialization.

Type I error at $\alpha = 0.05$ (MCSE 0.022) varies sharply across candidate test forms, motivating the choice of the primary class-aware moderation test used in the main manuscript:

Test	Rejection rate
lme `bm:Dbc` Wald t-test	0.000
lcmm gating-on-`bm` Wald z (unconditional)	**0.250**
lcmm within-class `bm:Dbc` Wald z	0.000
BIC selects ng=2 over ng=1	0.010
LRT chi-sq(df=4), naive reference	0.146
BIC-conditional gating procedure	0.000

The unconditional Wald-on-gating test rejects at five times the nominal rate. Diagnosis of the rejecting replicates shows that the empirical standard deviation of the gating coefficient across replicates (sd = 63.9) is roughly fifty times the median within-replicate Wald SE (0.86), and the empirical sd of the z-statistic is 1.47 against a nominal 1.0. The gating coefficient is centered at zero (median −0.10, sign balance 46/54) and shows no systematic bias; the inflation is driven by the conditional Wald SE under-estimating the true variability of $\hat{\beta}_{\mathrm{gating}}$ at trial-relevant N. This is a finite-sample identifiability artifact predicted by the[28] and[21] overextraction literature: when the data have no class structure but `hlme` is forced to fit `ng=2`, the optimizer latches onto random patterns to define a class boundary, and the resulting Wald SE does not absorb the labeling uncertainty.

The naive LRT against a chi-squared($df = 4$) reference is similarly anti-conservative (rejection 0.146), because the true LRT distribution at the `ng=1` boundary is a chi-bar-squared mixture rather than a chi-squared ([11], ch. 6). The bootstrapped LRT of[30] is the standard calibration but is computationally expensive at the per-fit budget of a class-aware simulation.

The pragmatic class-aware moderation test is therefore a BIC-conditional gating procedure: fit `ng=1` and `ng=2`, prefer `ng=2` only if BIC supports it ($\Delta_{\text{BIC}} > 0$, or more conservatively > 6 per Raftery 1995), and only then test the gating coefficient. At $N = 70$ under the null DGP, BIC selects `ng=2` in 1% of replicates, and zero of those replicates also reject the gating Wald, giving the conditional procedure a Type I error of 0 at the pilot's resolution. The conditional procedure is therefore usable as a primary moderation test at trial-relevant N.

The diagnostic distinction matters for the paper's argument. The proposition that 'class-aware analyses are not usable as primary at trial-relevant N' is too strong as stated: it applies to the unconditional Wald-on-gating test but not to the BIC-conditional procedure. The corrected proposition, carried into the main manuscript, is that the unconditional Wald test on a single mixture-model coefficient is unreliable at small N and requires a model-comparison wrapper (BIC or BLRT) to control Type I error.

Production-budget implication. Extrapolated to the pre-registered cell counts (Study 1 has 768 cells $\times$ 1,000 replicates = 768,000 fits per analysis arm), the class-aware arm of Study 1 is approximately 340 hours of serial CPU time at the pilot's mean fit time. Parallelisation across 8–16 cores reduces this to 22–44 hours per arm. Studies 2 and 3 require similar budgets; the Study 5 production run (5,000 replicates $\times$ three sample sizes $\times$ three class counts $\times$ three DGP families) is the largest single block at approximately 60 hours per arm on 8 cores. The full program is feasible on a single workstation over a week of background compute, with the effect-size tuning sub-study and Study 5 pre-flight running first to finalise effect-size axes and identifiability cuts before committing to Studies 1, 2, and 3.

Open question raised by the pre-flight pilot. With Type I error of the BIC-conditional procedure controlled at the null at $N = 70$, the next-most-pressing identifiability question was whether the BIC selection step has enough sensitivity to detect a real two-class DGP at trial-relevant N. The 240-replicate pilot reported in the main manuscript answers this directly: the BIC-conditional procedure is severely under-powered under the heterogeneous-random-slopes DGP at this sample size.

7 References

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